SUPPLY CHAIN ANALYTICS

Preliminary Project Review, Requirements Analysis, and Four-Week Implementation Plan

Project: Supply Chain Dataset Analysis

Prepared from the supplied project brief and dataset
Structure adapted from the provided Literature Review example

1. Introduction

Supply chain operations connect sourcing, production, inventory management, quality inspection, transportation, and delivery. Decisions across these activities can affect product availability, operating cost, delivery time, and the ability to meet demand. When information is reviewed in separate spreadsheets or through isolated metrics, decision-makers may find it difficult to identify where cost, lead time, stock, or quality issues deserve attention.

This project uses the provided Supply Chain dataset to explore operational and commercial indicators in a single analytical workflow. The intended outcome is a cleaned dataset, a documented set of business questions, and a Power BI dashboard that helps users compare products, suppliers, locations, production indicators, inspection outcomes, shipping methods, and costs.

This document follows the organization of the supplied PortfolioGenie Literature Review as a structural reference. The content has been rewritten for the Supply Chain Analytics project; PortfolioGenie itself is not part of this project. The review is preliminary and should be updated after the team completes data cleaning, validates measures, and confirms findings from the dashboard.

2. Review Method and Scope

The review is based on three project inputs: (1) the Supply Chain project brief, which describes a four-week workflow; (2) the supplied CSV dataset; and (3) the section structure of the example document. No claim is made that the dashboard or analysis has already been completed.

Scope of the project: inspect and clean the data; define decision-oriented analytical questions; build Power BI views to answer the questions; and prepare a final report and presentation describing the work, analysis, model development, and proposed deployment.

Evaluation criteria

- Data quality: completeness, consistency, valid data types, and row-level integrity.
- Decision usefulness: whether a metric can support a specific operational or managerial question.
- Coverage: product, inventory, supplier, production, quality, transportation, and cost dimensions.
- Interpretability: whether dashboard visuals and labels make comparisons easy to understand.
- Traceability: whether each KPI and conclusion can be linked to a defined field or calculation.
- Limitations: whether the dataset supports the proposed conclusion, and what cannot be inferred.

3. Dataset Overview

Initial file inspection found 101 rows and 24 columns. The file contains product-level and operational fields such as Product type, SKU, Price, Availability, Number of products sold, Revenue generated, Stock levels, Lead times, Order quantities, supplier and location fields, production volumes, manufacturing lead time, manufacturing costs, inspection results, defect rates, transportation modes, routes, and shipping costs/times.

The initial inspection identified 23 columns with missing values and 0 fully duplicated rows. These are preliminary profiling results, not final cleaning decisions. The team should inspect each missing value and suspicious record before choosing to correct, retain, label as unknown, or remove it.

Main field groups

- Product and commercial performance: Product type, SKU, Price, Number of products sold, Revenue generated, Customer demographics.
- Inventory and ordering: Availability, Stock levels, Order quantities, Lead times, Lead time.
- Supplier and production: Supplier name, Location, Production volumes, Manufacturing lead time, Manufacturing costs.
- Quality: Inspection results, Defect rates.
- Shipping and logistics: Shipping times, Shipping carriers, Shipping costs, Transportation modes, Routes, Costs.

Important data-model note: the dataset includes both “Lead times” and “Lead time”, and both “Shipping costs” and “Costs”. These similarly named fields must not be assumed to mean the same thing. Their definitions should be confirmed from the dataset documentation or project owner before creating combined KPIs.

4. Review of Analytical Tools

4.1 Microsoft Power BI

Role in this project: primary tool for interactive reporting and dashboard delivery, matching the project brief. Power Query can be used for transformation, the data model can organize fields for analysis, and DAX measures can calculate KPIs.

Strengths

- Supports interactive visuals and filtering across product, supplier, location, quality, and logistics dimensions.
- Combines data preparation, modelling, measures, and reporting in one workflow.
- Suitable for presenting operational KPIs to a non-technical decision-maker.

Considerations

- Visuals can mislead if measures, units, missing values, or field definitions are not validated first.
- A static dataset with no date field cannot support genuine time-series trends unless a valid date source is added.
- The team must distinguish sum, average, count, and weighted rate calculations where appropriate.

4.2 Microsoft Excel

Role in this project: an optional inspection and validation aid. Excel can help the team review sample records, compare totals, and manually check transformation outcomes before loading data into Power BI.

Strengths

- Accessible for spot checks, simple summaries, and collaboration with teammates who are less familiar with code.
- Useful for validating a small number of records and comparing calculated values.

Considerations

- Manual cleaning steps can be difficult to reproduce consistently unless they are documented.
- Large or repeated transformations are better managed through a repeatable query or script.

4.3 Python and pandas

Role in this project: an optional preprocessing and data-profiling tool, especially if the course deliverable explicitly asks for a preprocessing notebook. A notebook can document data types, missing-value checks, duplicate checks, and repeatable transformations.

Strengths

- Repeatable data-quality checks and clear records of the cleaning process.
- Convenient for profiling many columns and checking unexpected values.

Considerations

- The notebook should be understandable and reproducible; code should not replace an explanation of why each cleaning decision was made.
- If the required workflow is Power BI only, Python should be treated as a supporting tool rather than an unapproved substitute.

4.4 Tableau

Tableau is another business-intelligence tool capable of interactive visual analysis. It is included only as a comparison point, not as a proposed project dependency. Since the supplied brief names Power BI, the team should prioritize Power BI unless the supervisor changes the requirement.

5. Comparative Summary

Tool	Main role	Strength for this project	Limitation / decision
Power BI	Primary dashboard and model	Interactive reporting, Power Query, DAX	Use as the main delivery tool; validate KPIs and definitions.
Excel	Spot checks and simple validation	Easy record-level inspection	Optional support; document manual changes.
Python / pandas	Repeatable profiling and preprocessing notebook	Reproducible checks and transformations	Use if required or approved; explain decisions.
Tableau	Alternative BI tool	Interactive visual analysis	Not needed for the stated brief, which specifies Power BI.

Interpretation: the tools serve different purposes. Power BI is the primary reporting environment. Excel can support manual validation, while Python/pandas can support a reproducible preprocessing notebook if required. The team should avoid duplicating work across tools without a clear reason.

6. Identified Analytical Gaps and Opportunities

The dataset offers several opportunities for operational analysis, but its limitations should shape the questions and the wording of conclusions.

- Inventory visibility: compare Stock levels and Availability across products and product types; identify low-stock records for review. A low-stock threshold must be agreed by the team rather than invented after seeing the chart.
- Commercial comparison: compare Revenue generated and Number of products sold by Product type and SKU. Revenue is present as a field; do not assume it equals Price multiplied by units sold unless validated.
- Supplier and location comparison: summarize supplier-related lead-time, production, manufacturing-cost, and quality indicators where definitions and row granularity support a fair comparison.
- Quality monitoring: compare Inspection results and Defect rates by product type, supplier, or location. Confirm the unit and meaning of Defect rates before interpreting the values as percentages.
- Logistics cost and speed: compare Shipping times, Shipping costs, Transportation modes, Routes, and Shipping carriers. Do not label one carrier or route as “best” based on cost alone; include speed and data coverage.
- Operational efficiency: explore whether longer lead-time values coincide with higher costs, lower stock, or quality outcomes. Treat these as associations, not proof that one factor causes another.
- Data-quality transparency: include a record of missing fields, unknown categories, duplicate handling, and unresolved definitions so stakeholders can judge confidence in the results.

7. Suggested Differentiation and Initial Priorities

The project should be positioned as a decision-support dashboard, not simply a collection of charts. Its value is the connection between a clearly defined business question, a validated measure, a visual comparison, and a practical action or follow-up question.

Recommended priorities

Priority 1 — Profile the source data and document all quality issues before analysis.

Priority 2 — Agree on field definitions, units, and the meaning of similarly named columns.

Priority 3 — Create a short, prioritized list of business questions and map each question to fields, calculations, and visuals.

Priority 4 — Build a Power BI report with consistent labels, filters, and clearly defined KPIs.

Priority 5 — Validate totals and sample calculations independently; document assumptions and limitations.

Priority 6 — Present findings only after analysis is completed, then convert supported findings into recommendations.

8. Proposed Business Questions

The questions below are proposed for Week 2. The team should select the most useful ones after checking data quality and confirming metric definitions. They are questions to investigate, not findings already established.

Business question	Relevant fields	Possible analysis / visual	Decision use
Which product types generate the most recorded revenue and units sold?	Product type, Revenue generated, Number of products sold	Bar charts; side-by-side comparison	Understand product-category performance.
Which SKUs combine high sales with low stock levels?	SKU, Number of products sold, Stock levels, Availability	Scatter plot or table with conditional formatting	Prioritize inventory review.
How do supplier lead-time indicators differ?	Supplier name, Lead times, Lead time, Location	Supplier comparison bars and detail table	Identify suppliers or records for follow-up; confirm field definitions first.
How do manufacturing costs and production volumes vary by supplier or product type?	Manufacturing costs, Production volumes, Supplier name, Product type	Bar charts and scatter plot	Compare operational patterns; avoid causal claims.
Do inspection outcomes or defect rates vary across suppliers, product types, or locations?	Inspection results, Defect rates, Supplier name, Product type, Location	Stacked bars, average/median comparison, detail table	Focus quality checks where indicators warrant review.
How do shipping costs and shipping times compare by carrier and transport mode?	Shipping costs, Shipping times, Shipping carriers, Transportation modes	Grouped bars and scatter plot	Compare cost-speed trade-offs.
Which routes have higher recorded logistics costs or shipping times?	Routes, Costs, Shipping costs, Shipping times	Route comparison chart	Identify routes that may need investigation; clarify cost fields.

Business question	Relevant fields	Possible analysis / visual	Decision use
Is stock availability aligned with recorded sales volume?	Availability, Stock levels, Number of products sold, SKU	Scatter plot or matrix	Spot records that may require replenishment or data validation.

Prioritization recommendation: start with product performance, inventory, supplier lead-time, quality, and shipping cost-versus-time. Keep the final question list short enough that every question can be answered clearly in the dashboard.

9. Proposed Dashboard Structure

Page 1 — Executive Overview

- Cards for total recorded revenue, total units sold, average stock level, average shipping time, and average shipping cost, provided definitions are validated.
- Breakdowns by product type and location.
- A concise “items to review” section based on transparent thresholds chosen by the team.

Page 2 — Product and Inventory

- Revenue and units sold by product type and SKU.
- Stock levels and availability by SKU.
- A table for products that meet the agreed low-stock / high-sales criteria.

Page 3 — Supplier and Production

- Supplier comparison using confirmed lead-time fields.
- Production volumes and manufacturing costs by supplier or product type.
- Location slicer and a record-level detail table.

Page 4 — Quality and Logistics

- Inspection results distribution and defect-rate comparison.
- Shipping time and shipping cost comparison by carrier and transportation mode.
- Route-level comparison with a note distinguishing Shipping costs from Costs.

The team may combine pages if a smaller report is more readable. Do not add a time trend chart unless a valid date field is available. Avoid displaying every possible metric on one page.

10. Requirements Analysis

10.1 Stakeholders and their needs

Project team / data analysts

- Need a repeatable cleaning workflow and a clear record of assumptions.
- Need agreed metric definitions and a model that supports accurate comparisons.
- Need a manageable list of analytical questions and a clear division of work.

Operations or supply-chain decision-maker

- Needs to compare product, inventory, supplier, production, quality, and logistics indicators.
- Needs to identify records or categories that warrant follow-up without treating correlation as causation.
- Needs clear labels, filters, definitions, and visible caveats for incomplete data.

Project supervisor / evaluator

- Needs evidence that the team followed the four-week workflow and delivered the expected outputs.
- Needs a final report and presentation explaining the process, analysis, model, findings, limitations, and proposed deployment.

10.2 Functional requirements

FR1 — The workflow shall load the supplied CSV and record its initial row count, column count, and data types.

FR2 — The cleaning process shall identify missing values, duplicates, invalid types, inconsistent categories, and suspicious records.

FR3 — Each cleaning decision shall be documented, including whether values were corrected, retained, categorized, or removed.

FR4 — The team shall produce a prioritized list of business questions, with each question mapped to dataset fields and a proposed analysis.

FR5 — The Power BI report shall provide interactive comparisons using relevant product, supplier, location, quality, and logistics fields.

FR6 — Measures and visuals shall have clear names, units, and definitions.

FR7 — The final report and presentation shall summarize data preparation, analysis, model development, results, limitations, and proposed deployment.

10.3 Non-functional requirements

- Accuracy: calculations and visuals must be checked against the source data.
- Clarity: titles, axes, units, and filters should be understandable to a non-technical audience.
- Reproducibility: cleaning steps and important assumptions should be recorded.
- Usability: pages should have consistent formatting and avoid unnecessary visual clutter.
- Traceability: conclusions should be linked to a measure and the data supporting it.

11. User Stories and Core Use Cases

11.1 User stories

User Story 1: As a data analyst, I want to inspect missing values and duplicates so that I can make informed and documented cleaning decisions.

User Story 2: As a project teammate, I want each business question mapped to fields and a visual so that the analysis stays focused on decision needs.

User Story 3: As an operations decision-maker, I want to compare stock levels with recorded sales so that I can identify products requiring inventory review.

User Story 4: As a supply-chain manager, I want to compare shipping cost and time by carrier and mode so that I can review trade-offs.

User Story 5: As a supervisor, I want the final report to explain the workflow, assumptions, results, and limitations so that I can evaluate the project consistently.

11.2 Core use cases

- Data profiling and cleaning: inspect the source, document quality issues, apply approved transformations, and export a cleaned dataset.
- Question definition: agree on the business questions, required fields, measures, and visual formats.
- Dashboard exploration: use slicers and charts to compare categories and inspect details.
- Validation: compare dashboard values with source-level checks and revise calculations if needed.
- Final communication: present only supported findings, recommendations, and known limitations.

12. Four-Week Implementation Plan and Deliverables

Week	Main work	Expected deliverable
Week 1 — Data cleaning and preprocessing	Inspect schema; review missing values and duplicates; validate data types and categories; document transformations; confirm ambiguous field definitions.	Cleaned dataset and preprocessing notebook / documented transformation workflow, as required by the brief.
Week 2 — Business questions	Turn the dataset into decision-oriented questions; prioritize questions; map fields, calculations, and intended visuals; review feasibility.	A concise list of analytical questions with their purpose and required fields.
Week 3 — Power BI dashboard	Build the data model and measures; create pages and filters; validate KPIs; check readability and consistency.	Interactive Power BI dashboard answering the selected questions.
Week 4 — Final report and presentation	Summarize the problem, workflow, data preparation, model, analysis, results, recommendations, limitations, and deployment approach.	Final written report and presentation.

Deliverable status note: this document proposes the structure and plan. It does not claim that cleaning, dashboard development, or final analysis has already been completed.

13. Validation and Quality Checklist

- Confirm whether the extra final partial row in the CSV is a valid record or a malformed line before deciding how to handle it.
- Check that SKU is unique where expected and investigate blank or repeated identifiers.
- Review missing values by column and choose a justified treatment for each field.
- Confirm numeric types, units, and plausible ranges for prices, costs, stock, quantities, lead times, and defect rates.
- Clarify whether “Lead times” and “Lead time” are distinct measures.
- Clarify whether “Shipping costs” and “Costs” represent different cost concepts.
- Check whether “Revenue generated” can be reconciled with price and units sold; do not assume the formula without validation.
- Check that defect-rate units are understood before formatting them as percentages.
- Validate DAX totals against source-level calculations.
- Check all slicers, cross-filtering, titles, labels, units, and totals.
- Ensure final recommendations follow from the analysis and are phrased with appropriate caution.

14. Limitations and Risks

- No date column is apparent in the supplied field list; time-based trends and forecasting are therefore out of scope unless another valid date source is provided.
- The dataset appears to contain a small number of records relative to a real supply-chain operation, so conclusions may not generalize beyond the supplied data.
- Missing, unknown, or malformed values may affect comparisons; data cleaning decisions must be documented.
- Several field names are ambiguous or similar. Incorrectly combining them could produce misleading KPIs.
- The dataset is observational. Relationships between lead time, cost, stock, and quality should not be presented as causal effects.
- No business targets or service-level thresholds were provided. Thresholds such as low stock or acceptable lead time must be agreed with the supervisor or clearly labeled as exploratory assumptions.

15. Conclusion

The Supply Chain Dataset Analysis project can turn a multi-area dataset into a structured decision-support report by combining careful data preparation, focused business questions, and an interactive Power BI dashboard. The most important early task is not chart design; it is confirming data quality and metric definitions so that comparisons are trustworthy.

The proposed workflow follows the four-week project brief: clean and document the data, define the questions, build a dashboard that answers them, and communicate the completed work in a final report and presentation. The final conclusions and recommendations should be written only after the team has validated its analysis.

References and Project Inputs

Project inputs used for this draft:

- Supply Chain project brief provided in the conversation (four-week workflow and expected deliverables).
- supply_chain_data(1).csv — dataset supplied for the project.
- 2_Literature_Review.pdf — used as a structural reference for organization and section types only; its PortfolioGenie-specific content has not been reused as Supply Chain facts.

Tool descriptions in this document are high-level project-planning context. The team should consult current official documentation if it needs exact instructions for Power BI, Power Query, DAX, Excel, or pandas.